

Data Visualization Analysis Report: Sample Superstore Dataset

Data Analysis Submission

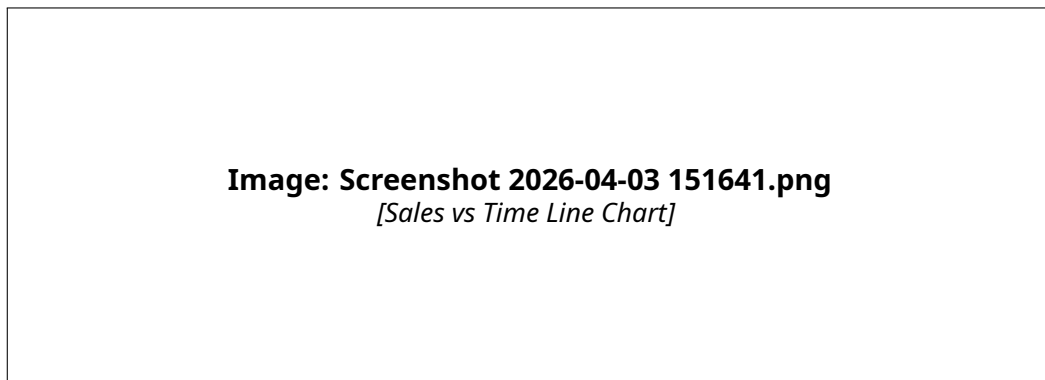
April 3, 2026

Introduction

This report provides a comprehensive analysis of the Sample Superstore dataset through various data visualization techniques. Each visualization aims to uncover specific business insights regarding sales performance, profitability, regional contributions, and operational trends. The following analysis is based on data extracted from the 2014–2017 fiscal periods.

Visualizations and Interpretations

1. Line Chart: Sales Over Time (2014–2017)



Interpretation: This chart tracks sales across three categories. After stagnation in 2015, growth spiked. **Technology** led by 2017 with approximately \$271,731 in sales, suggesting high demand for premium electronics.

2. Scatter Plot: Relationship Between Sales and Profit

Interpretation: A positive correlation exists, but **Furniture in the Central region** is a major outlier, showing losses despite sales. Profit is healthiest in the **East and West** for Technology.

3. Bar Chart: Categorical and Sub-category Analysis

Interpretation: **Phones** drive volume, but **Copiers** drive high-margin profit. **Tables** and **Book-cases** are identified as loss-making sub-categories requiring urgent strategy adjustment.

Image: Screenshot 2026-04-03 151652.png
[Sales vs Profit Scatter Plot]

Image: Screenshot 2026-04-03 151700.png
[Categorical Bar Chart]

4. Pie Chart: Region-wise Contribution

Image: Screenshot 2026-04-03 151711.png
[Region-wise Pie Chart]

Interpretation: The **West** (31.58%) and **East** (29.55%) dominate total revenue, while the **South** remains an under-penetrated market with significant room for growth.

5. Trend Line: Discount Patterns

Interpretation: Discounting has increased steadily, particularly in **Office Supplies**. This correlates with the sales growth but threatens the average profit per unit.

6. Histogram: Profit Distribution

Interpretation: Most transactions cluster near zero profit. The histogram highlights significant volatility, with occasional large losses impacting overall category health.

Image: Screenshot 2026-04-03 151719.png
[Discount Trend Line]

Image: Screenshot 2026-04-03 151727.png
[Profit Distribution Histogram]

Conclusion

The Superstore is growing, but profitability is inconsistent. Success in the West/East Technology segments should be used as a model to fix issues in the Central Furniture segment.